Introduction to Robotics: Homework II

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Introduction

The homework involves determining the transformation matrices and robot joint positions associated with a robotic manipulation task. Specifically, you are required to identify the transformation matrices for all objects in the environment, the relative transformation matrix between the manipulator's end-effector and the object being manipulated, and the transformation matrix for the end-effector that enables the manipulated object to be precisely inserted into a designated hole. Once all the transformation matrices are determined, you are required to calculate using inverse kinematics the joint positions of the manipulator in order to a) grasp the cylinder, and b) place the cylinder on top of the target hole.

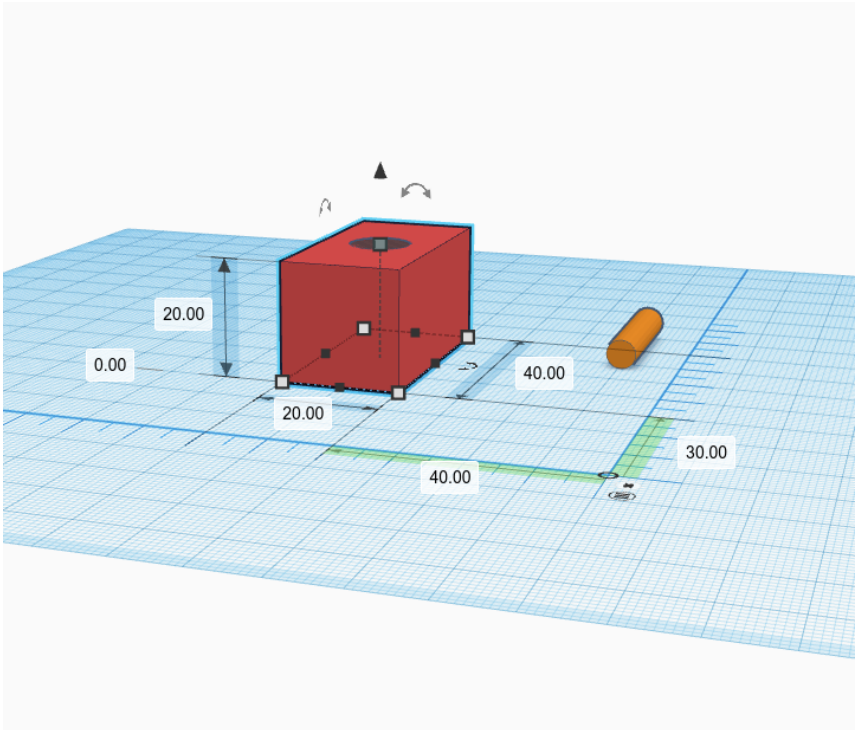

Figure 1: Visualization of the manipulation scenario. **Measurements are in cm.**

1 Description

The setup includes a Franka Panda manipulator equipped with a gripper (Fig. 2), a box featuring a cylindrical hole at its center (represented as the red box in Fig. 1), and a cylinder (shown as the orange cylinder in Fig. 1) that the manipulator must handle and insert into the hole. The Franka Panda is a versatile, 7-DOF robotic arm designed for precise manipulation tasks, widely used in research and industrial applications due to its sensitivity and ease of control. **The kinematic structure of the robot is given in Fig. 3. The transformation from the frame x_F to**

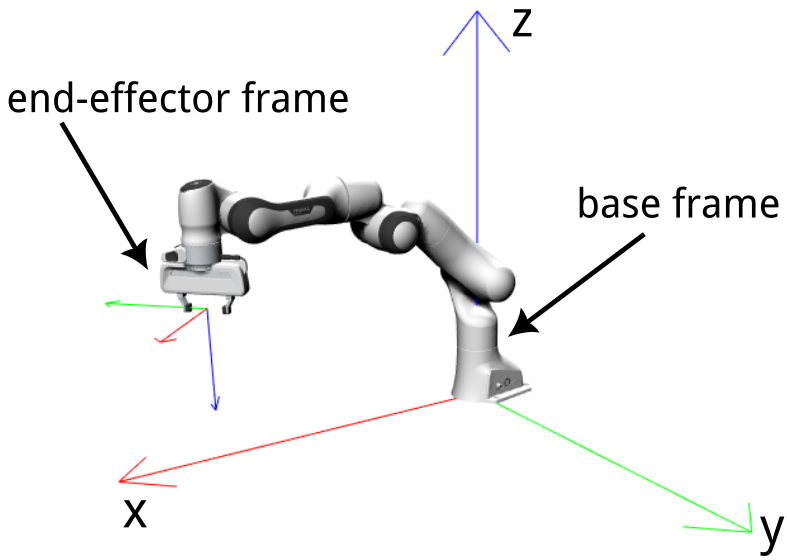

Figure 2: Visualization of the Franka Panda manipulator.
the end-effector is given by a rotation of $-\frac{\pi}{4}$ around the z-axis and a translation of $0.103\,m$ in the z-axis.

2 Task

Given Fig. 4, Fig. 5, Fig. 6, Fig. 1, Fig. 7, Fig. 8, Fig. 2, and Fig. 3:

1. Choose the location of the world frame and assign a fixed body frame to each object;
2. Select an appropriate transformation matrix for the base frame of the manipulator with respect to the world frame, ensuring it can perform the manipulation task;
3. Determine the relative transformation matrix between the manipulator's end-effector and the cylinder, allowing the manipulator to grasp the cylinder (aka, what should be the orientation and position of the cylinder with respect to the end-effector such that it can be grasped);
4. Calculate the world frame transformation matrix of the cylinder such that, when released, it will be positioned to drop precisely into the hole in the box;
5. Using inverse kinematics calculate the joint positions of the arm such that the end-effector is able to grasp the cylinder (verify your result with forward kinematics);
6. Using inverse kinematics calculate the joint positions of the manipulator such that the cylinder (grasped by the arm) is positioned in the "placing" position above the hole (verify your result with forward kinematics);
7. Visualize the results using the programming language of your choice.

Guidelines:

- You have to use the Product of Exponentials for the solving the forward kinematics problem;
- You have to solve the inverse kinematics problem using the Newton-Raphson algorithm;
- In order to solve the inverse kinematics problem you have to be able to compute the Jacobians of the manipulator (compute the geometric Jacobians);
- Your inverse kinematics solver should be generic and applicable to any desired end-effector pose.

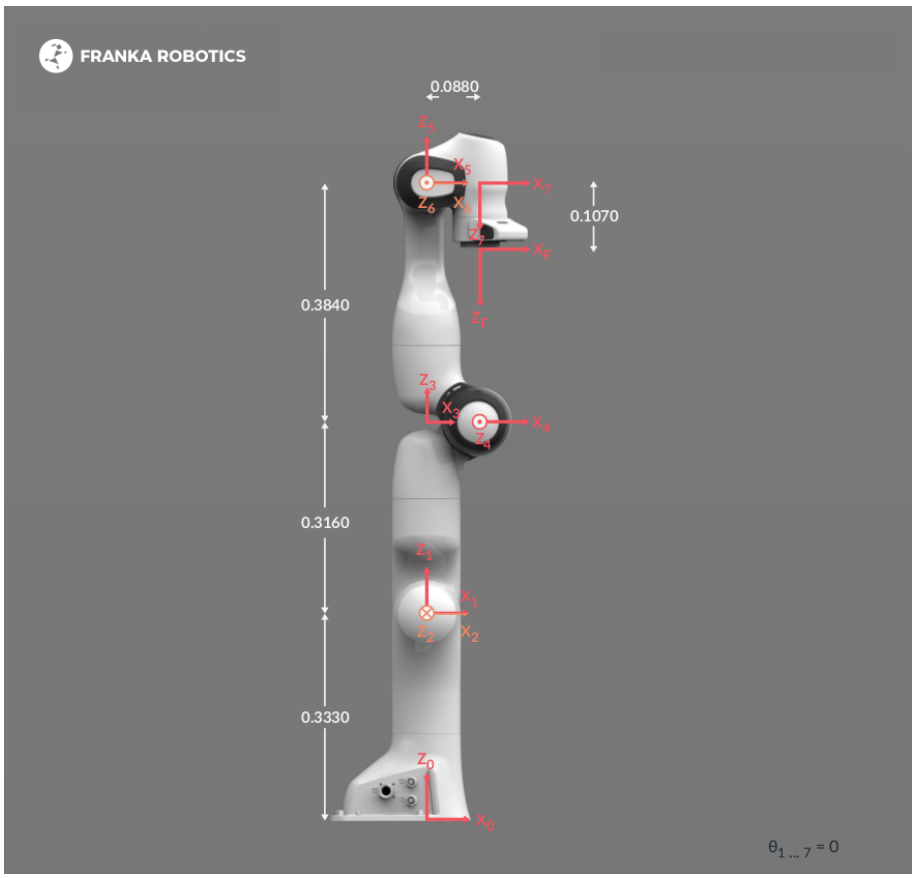

Figure 3: Franka Panda manipulator kinematic structure. **Measurements are in meters.**

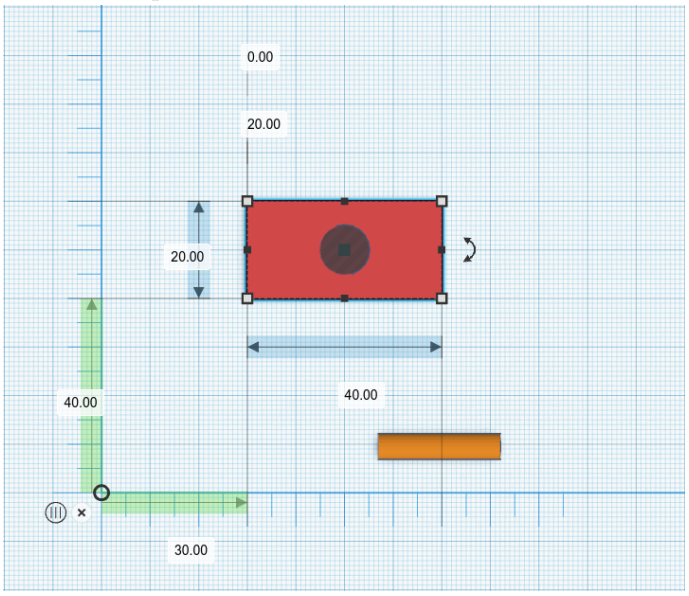

Figure 4: Scenario measurements. **Measurements are in cm.**

3 Deliverables

- Detailed report;
- Code for visualization and containing the transformation matrices.

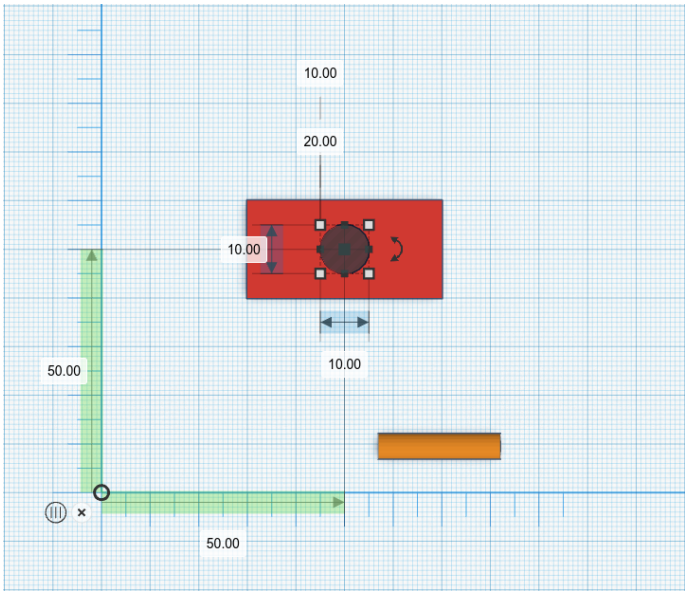

Figure 5: Scenario measurements. **Measurements are in cm.**

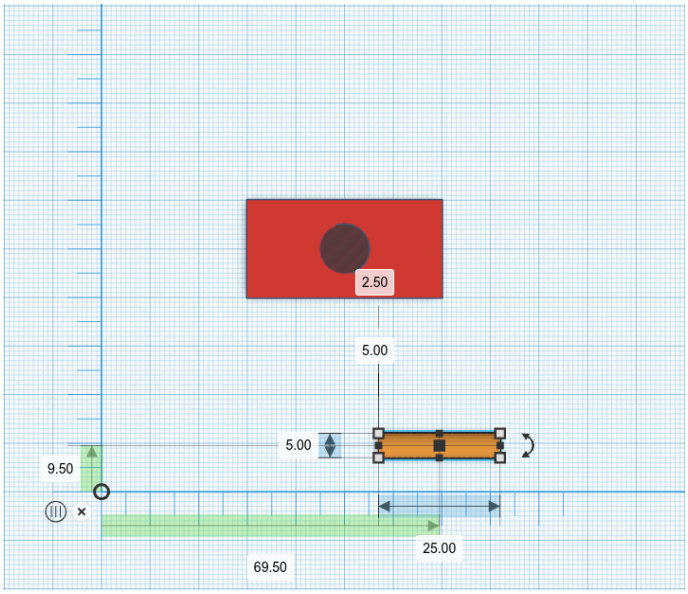

Figure 6: Scenario measurements. **Measurements are in cm.**

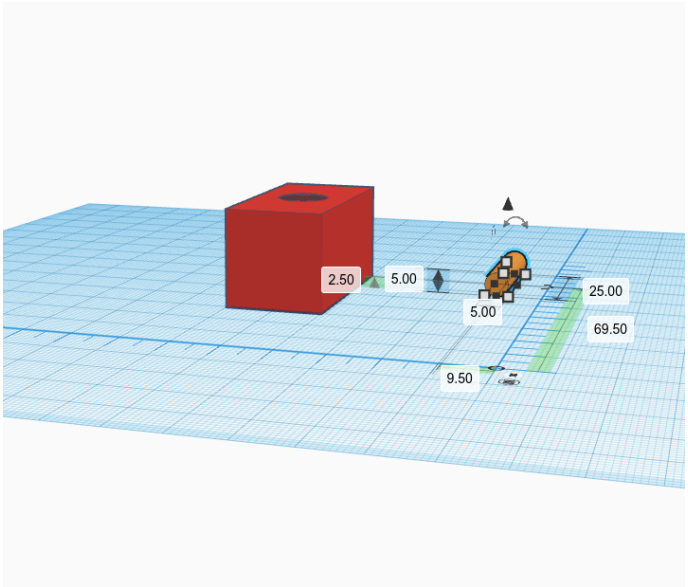

Figure 7: Scenario measurements. **Measurements are in cm.**

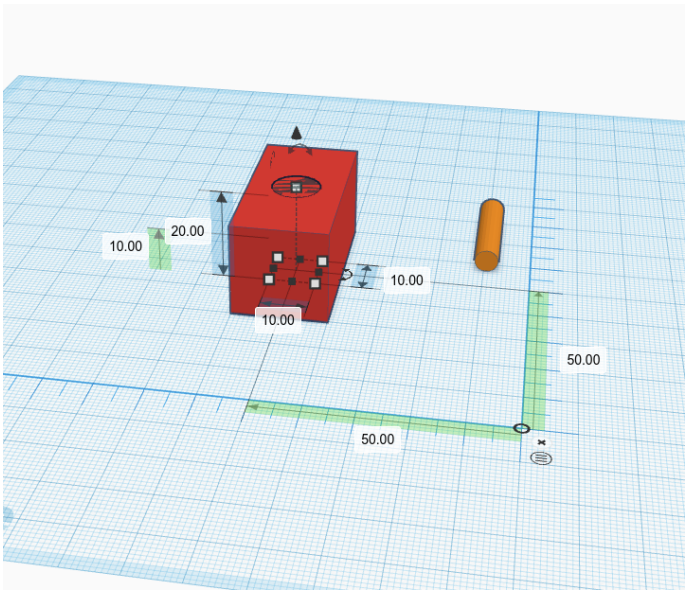

Figure 8: Scenario measurements. **Measurements are in cm.**